Mohamed Alaaeldin Hassan Hassan

Mechanical R&D / Design Engineer / Production Engineer | Product Development ·FMEA ·Continuous Improvement

Portfolio

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PROFESSIONAL SUMMARY

Mechanical Engineer with a Bachelor's in Mechanical Production & Design Engineering and six SolidWorks certifications spanning part/assembly design, sheet metal, weldments, drawings, and simulation, alongside FMEA Specialist and Lean Six Sigma Yellow Belt credentials. Promoted to System Improvement Team Leader at SEWS-E, leading a role spanning two areas of process improvement — mechanical design of jigs, fixtures, and production tooling to eliminate recurring defects, and FMEA (having progressed from FMEA Team Leader to FMEA Champion within that activity). Led the mechanical design of a Smart Recycled Plastic Pelletizing Machine as graduation project team leader, from concept through CAD modeling, material selection, and manufacturing preparation, securing support from three external organizations. Brings hands-on exposure to digital fabrication instruction, hydraulic and manufacturing systems, and structured quality-tool methodology.

WORK EXPERIENCE

R&D Engineer | Acrow Misr **Oct 2026 – Present**

Research & Development · Mechanical Engineering

Applying mechanical engineering and design knowledge to support research, product development, and practical engineering solutions.

System Improvement (R&D) Team Leader — Process Development & FMEA — SEWS-E **Oct 2025–Sept 2026**

Promoted to System Improvement Team Leader (Sep 2026) — role spans FMEA (risk analysis) and Mechanical Design (jigs, fixtures & tooling). Within the FMEA activity, took on increasing responsibility: FMEA Team Leader (Feb 2026) → FMEA Champion (May 2026).

- **Mechanical Design:** Design and develop jigs, fixtures, and production tooling to support manufacturing operations, engineering new tools to eliminate recurring defects and prevent repeat customer concerns.
- **FMEA:** Rebuilt and reactivated a dormant FMEA system, designing structured workflows and follow-up controls to restore risk-tracking and process accountability across engineering teams.
- **FMEA:** Led FMEA reviews to identify potential failure modes in production processes and drive corrective/preventive actions, applying PFMEA/RFMEA methodology.
- Promoted to System Improvement Team Leader (Sep 2026) based on project leadership and demonstrated impact, after taking on progressively greater FMEA responsibility (Team Leader → Champion) within the role.

SolidWorks & Digital Fabrication Instructor — ITIDA, Knowledge City, New Administrative Capital **Aug 2025**

- Instructed trainees in SolidWorks CAD fundamentals and digital fabrication workflows, translating design and manufacturing concepts into practical, hands-on sessions.

PROJECTS

Smart Recycled Plastic Pelletizing Machine — Graduation Project **2024 – 2025**

Team Leader & Lead Mechanical Designer

- Defined the design concept and led material selection for a machine to process recycled plastic into pellets, with sustainability as a core design driver.
- Led CAD modeling and simulation of the full assembly in SolidWorks, coordinating design decisions across subsystems with the project team.
- Directed the project from concept through design toward a manufacturable, buildable solution, working with the team on practical build considerations.
- Secured funding/support from three external organizations — Promech, Sector B5, and Egypt Makes.
- Graduation project rated “Excellent” by the evaluation committee.

FMEA System Revival & Continuous Improvement — SEWS-E **2026**

- Rebuilt a stagnant FMEA program from the ground up, embedding structured workflows and follow-up accountability into daily engineering operations.
- Recognized for leadership and measurable project impact with a promotion from FMEA Leader to FMEA Champion.

EDUCATION

Bachelor of Mechanical Production & Design Engineering — Helwan University, Egypt **2025**

- Graduated with Grade 72% (GPA 2.7/4.0)
- Graduation project rated “Excellent”

CERTIFICATIONS

- CSWP — SolidWorks Professional (Jan 2025)
- CSWA — SolidWorks Associate (Jul 2024)
- CSWPA-DT — Drawing Tools Professional (Apr 2025)
- CSWPA-WD — Weldments Professional (Apr 2025)
- CSWP-SM — Sheet Metal Professional (Jan 2025)
- CSWA-S — Simulation Associate (Sep 2024)
- FMEA Specialist — AIGPE, CPD Accredited (Jan 2026)
- Lean Six Sigma Yellow Belt — AIGPE, CPD Accredited (Sep 2025)

COURSES

- 7 Quality Tools — AmCham Egypt (Jun 2026) — Team ranked 1st among all teams in the course's final project
- CAMWORKS Turning — Net3lem.com (Sep 2025)
- Geomagic Design X — Net3lem.com (Aug 2025)
- CATIA Training Course — ProMech Group (Aug 2023)
- Hydraulic Course — Udemy (Jul 2023)
- CNC Turning Course — Faculty of Engineering, Helwan University (Feb 2022)

SKILLS

CAD & Mechanical Design: SolidWorks, CATIA, AutoCAD
Simulation & Manufacturing Prep: SolidWorks Simulation, CAMWORKS, Geomagic Design X
Quality & Continuous Improvement: FMEA (PFMEA/RFMEA), Lean Six Sigma, Minitab
Digital Fabrication & Prototyping: 3D Printing, Laser Cutting, Turning CAMWORKS,CNC Router operating, Rapid Prototyping
Visualization & Documentation: KeyShot, Photoshop CC, Microsoft Office

INTERNSHIPS

- The Arab Contractors — Engines (2022)
- Arab Organization for Industrialization — Aircraft Maintenance & NDT Training (2022)
- Yasser Fahmy Co. (authorized distributor: Bosch Rexroth, HYDAC Group, Emerson) — Maintenance (2023)
- EGYPTAIR Training Academy — Maintenance & NDT Training (2023)